Bank Marketing Campaign Effectiveness & Customer Conversion Analytics

An End-to-End Data Analytics Project for Improving Marketing Campaign Conversion

Dataset: UCI Machine Learning Repository — Bank Marketing (bank-additional-full.csv)
Source: https://archive.ics.uci.edu/dataset/222/bank+marketing
41,188 records · 20 input attributes · target variable: y (term deposit subscription)

1. Business Problem

A bank conducts direct (phone-based) marketing campaigns to encourage customers to subscribe to a term deposit. The bank wants to understand which customers are more likely to subscribe, which campaign characteristics drive success, which segments to prioritize, and how to improve overall campaign conversion.

Business question: Which customer and campaign characteristics are associated with successful term-deposit subscriptions, and how can the bank improve marketing campaign conversion?

2. Dataset

The UCI Bank Marketing dataset (bank-additional-full.csv) was used, containing 41,188 customer records and 20 input attributes across three groups: customer attributes (age, job, marital status, education, default, housing, loan), campaign/contact attributes (contact type, month, day of week, duration, number of contacts, days since previous contact, previous contacts, previous outcome), and economic context attributes (employment variation rate, consumer price index, consumer confidence index, euribor 3-month rate, number of employees). The target variable, y, indicates whether the customer subscribed to a term deposit.

Original source: S. Moro, P. Cortez and P. Rita, "A Data-Driven Approach to Predict the Success of Bank Telemarketing," Decision Support Systems, Elsevier, 62:22-31, 2014. The raw dataset was downloaded unmodified and is not claimed as original work of this project.

3. Data Cleaning

- 12 exact duplicate rows were identified and removed.
- No NaN-type missing values exist, but several categorical columns use the string "unknown" as a documented placeholder (e.g. default: 8,596 rows, education: 1,730 rows). These were kept as their own category rather than dropped or imputed, since "unknown" default status, for example, is itself informative.
- The pdays sentinel value of 999 ("never previously contacted") was converted into a readable was_previously_contacted flag.
- The target y was converted into a binary numeric column for analysis and modeling.

4. Exploratory Data Analysis — Key Observations

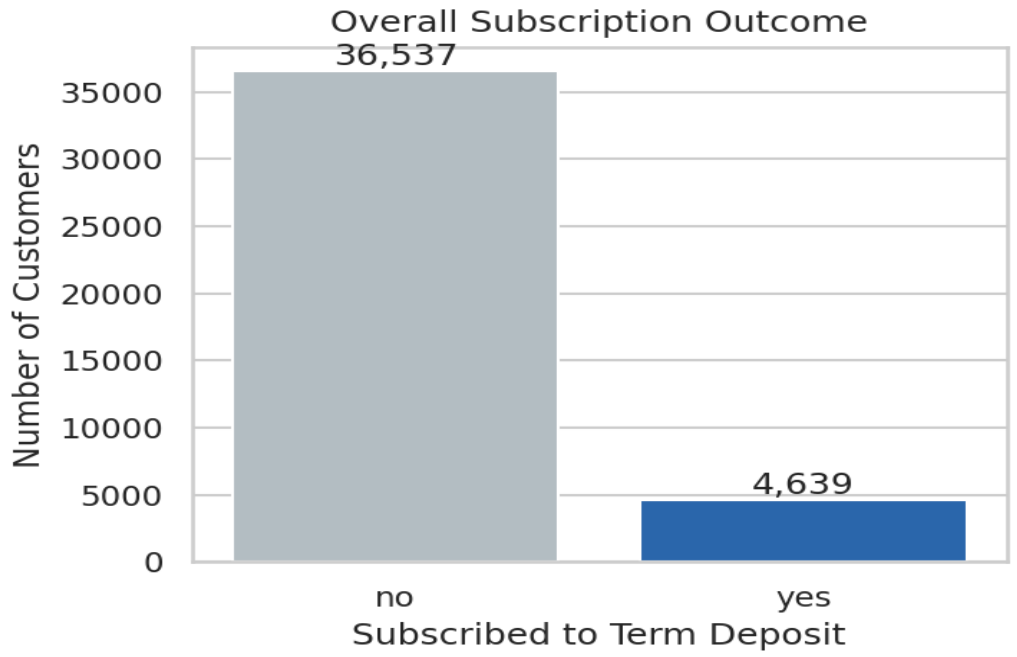


Figure 1. Overall subscription outcome — 41,176 customers after deduplication, ~11.3% converted.

4.1 Customer segments

Subscription rate varies substantially by occupation and age. Students (~31%) and retirees (~25%) convert far above the campaign average of 11.3%, while blue-collar (~7%) and services (~8%) workers convert below it.

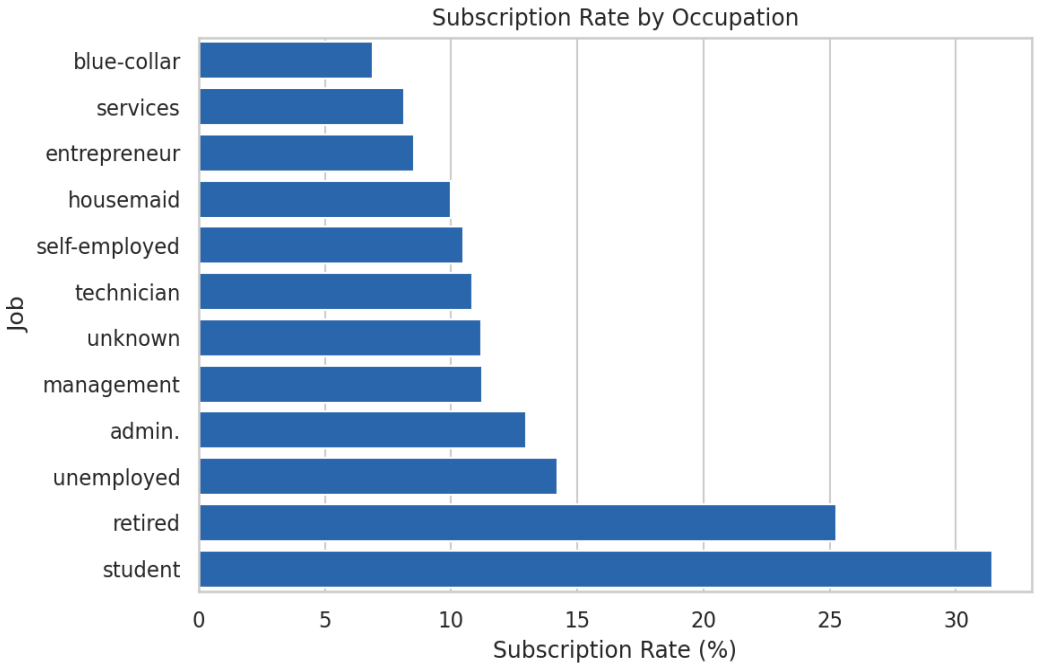


Figure 2. Subscription rate by occupation.

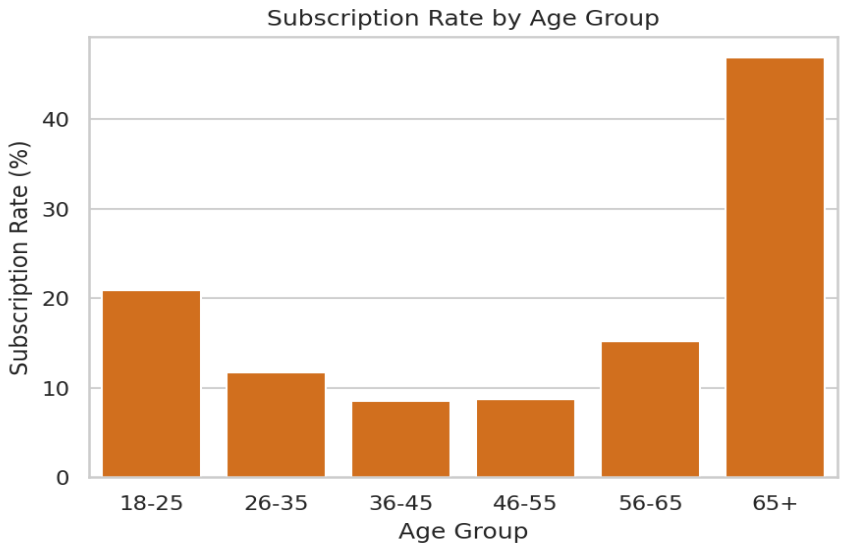


Figure 3. Subscription rate by age group — a clear U-shape, with the youngest and oldest customers converting best.

4.2 Contact channel and timing

Cellular contact converts at ~14.7% versus ~5.2% for telephone — nearly a 3x difference. Conversion also varies strongly by month: March, September, October and December convert above 40%, while May — the heaviest calling month in the dataset — converts at only ~6.4%.

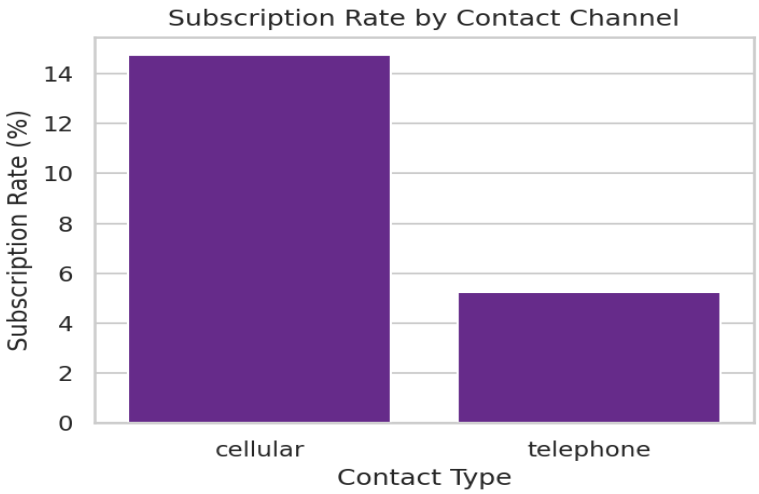


Figure 4. Subscription rate by contact channel.

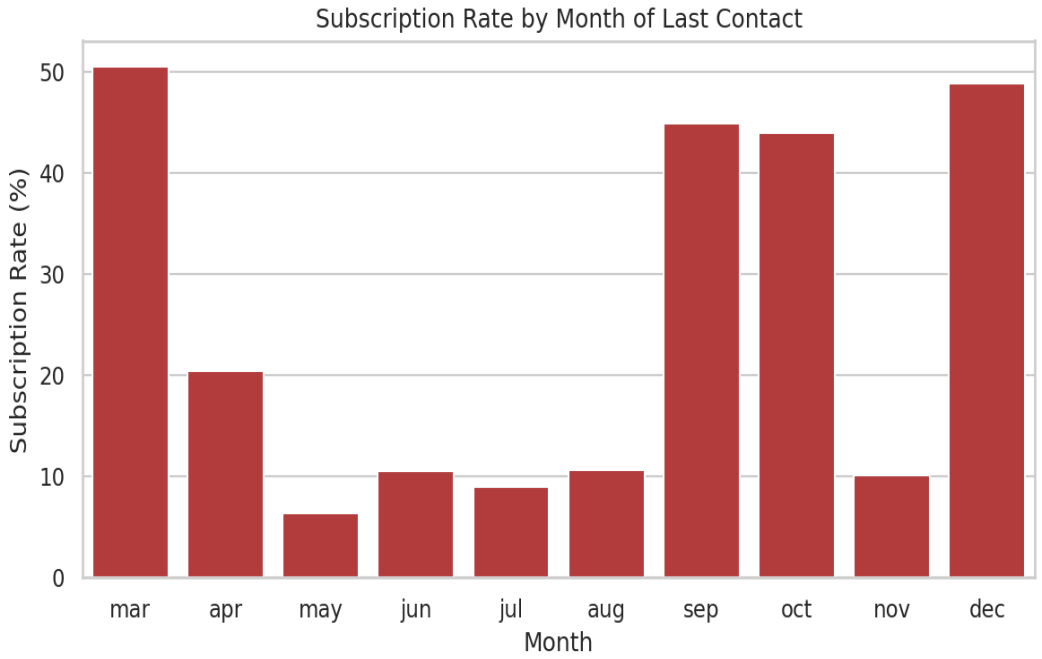


Figure 5. Subscription rate by month of last contact.

4.3 Contact frequency and campaign history

Customers who ultimately subscribed were contacted fewer times on average (~2.1 contacts) than those who did not (~2.6 contacts), suggesting diminishing returns to repeated calling within a campaign. Prior campaign outcome is the single strongest behavioral signal: customers with a previously successful outcome convert at ~65%, versus ~9% for customers never previously contacted.

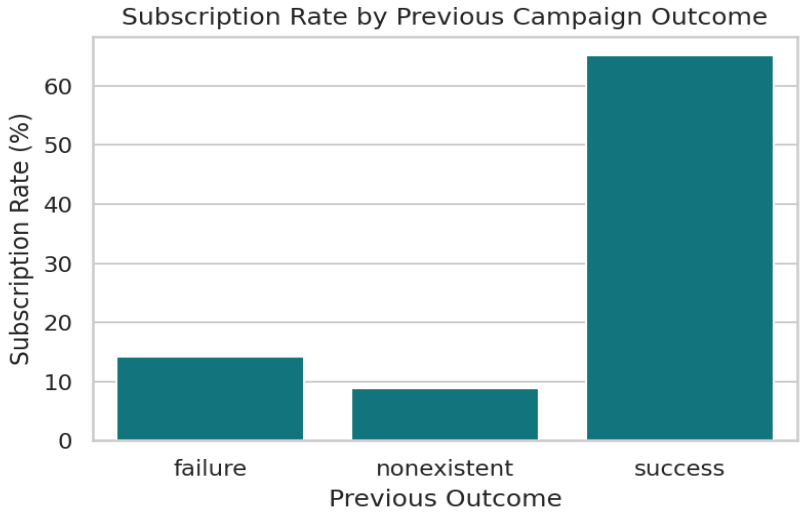


Figure 6. Subscription rate by previous campaign outcome.

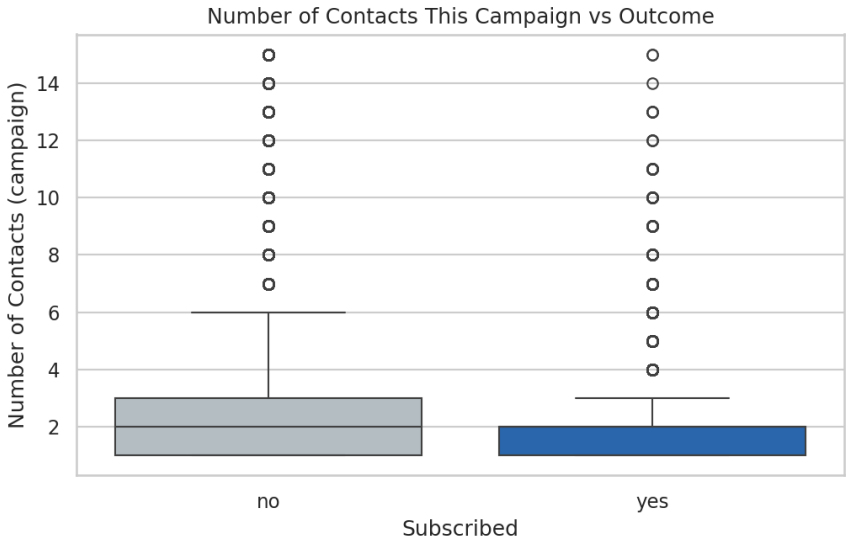


Figure 7. Number of contacts this campaign, by outcome.

4.4 Macroeconomic context

Numeric correlations show subscription is associated with macroeconomic conditions: average euribor 3-month rate is 2.12 among subscribers versus 3.81 among non-subscribers, indicating customers are more receptive to term deposits when interest-rate/employment conditions are comparatively cooler.

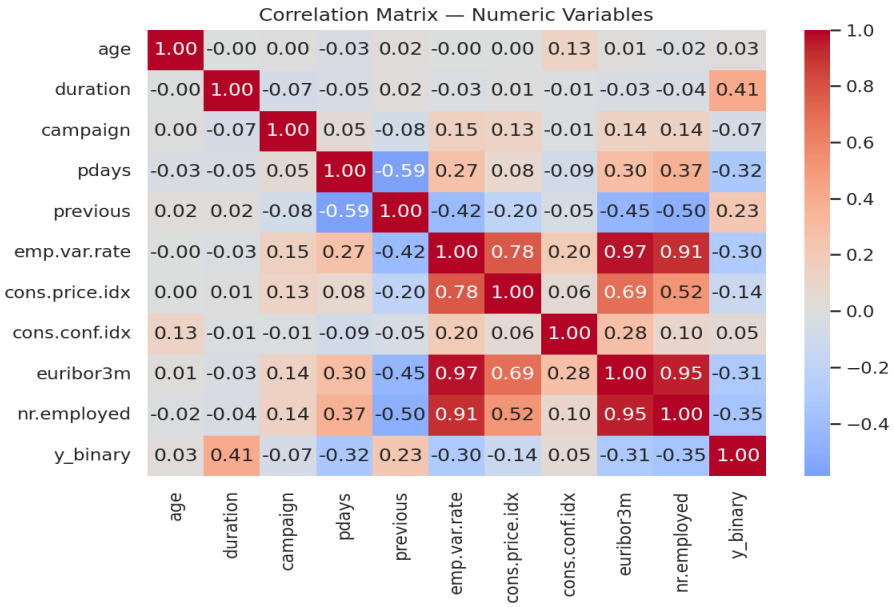


Figure 8. Correlation matrix of numeric variables.

5. Insights

- Segment targeting matters more than call volume — students and retirees are a small share of the customer base but convert at 2-3x the campaign average.
- Channel choice is a controllable lever — cellular contact strongly outperforms telephone for comparable effort.
- Timing is a controllable lever — campaigns run outside the heaviest calling months convert substantially better.
- There are diminishing (and likely negative) returns to contacting the same customer more than 2-3 times in one campaign.
- Past customer behavior is the strongest available signal — a previously successful customer is roughly 7x more likely to convert than a cold contact.
- Macroeconomic conditions shape overall appetite for term deposits, independent of how well any single campaign is executed.

6. Risk, Opportunity, and Action

Area	Risk	Opportunity	Action
Over-calling	Falling returns and customer fatigue beyond 3 contacts	Free up agent capacity	Cap standard campaigns at 2-3 contacts per customer
Channel mix	Telephone-only wastes agent time (<6% rate)	Cellular converts ~3x better	Prioritize cellular contact where a valid mobile number exists
Segment targeting	Undifferentiated lists dilute effort on low-converting groups	Students & retirees convert 2-3x average	Sequence call lists by expected conversion, not exclude segments
Seasonality	Heavy May-weighted calling wastes volume	Spring/autumn months convert far better	Shift more campaign volume to Mar/Sep/Oct/Dec where feasible
Re-engagement	Treating every contact as cold ignores a strong signal	Past successes convert at ~65%	Build a warm re-engagement list ahead of new campaigns

7. Predictive Analysis

Predictive analytics is appropriate here because the bank's practical need is to prioritize which customers to call next — a binary classification problem. A **Logistic Regression** model was chosen for its simplicity, speed, and interpretability, consistent with this project's goal of avoiding unnecessary complexity.

Leakage control: the duration field was excluded from the model, since it is only known after a call has already taken place and would not be available at the point the bank decides whether to call a customer.

Class imbalance: handled using class_weight="balanced" rather than more complex resampling methods, in keeping with the project's simplicity requirement.

7.1 Model performance (held-out 20% test set)

Metric	Score
Accuracy	0.83
Precision	0.36
Recall	0.65
F1-score	0.46
ROC AUC	0.80

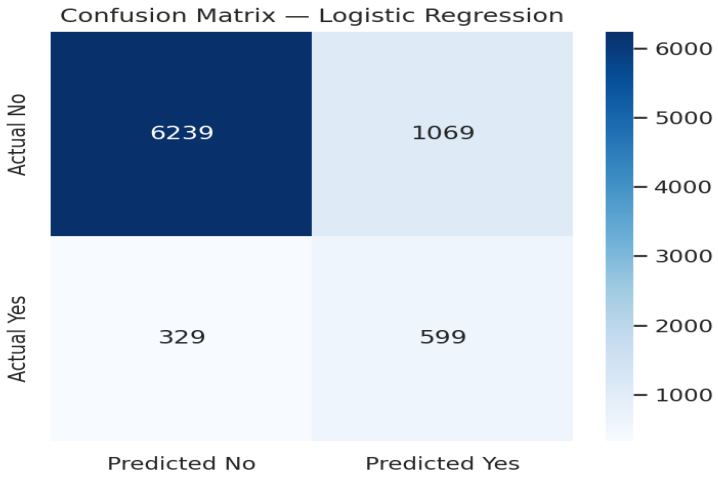


Figure 9. Confusion matrix on the held-out test set.

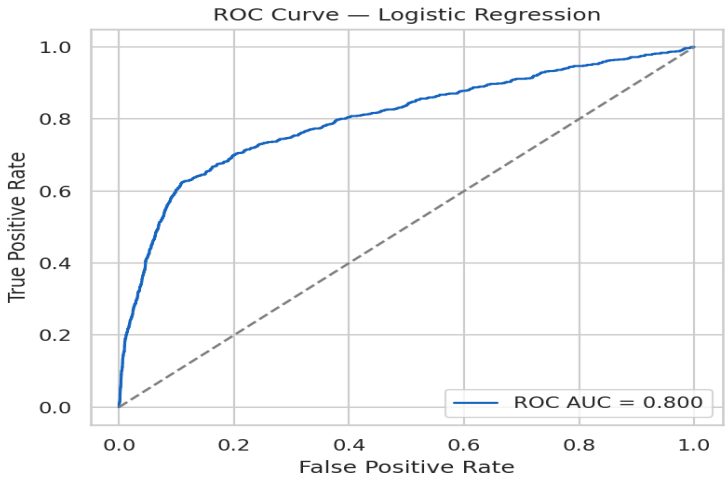


Figure 10. ROC curve (AUC = 0.80).

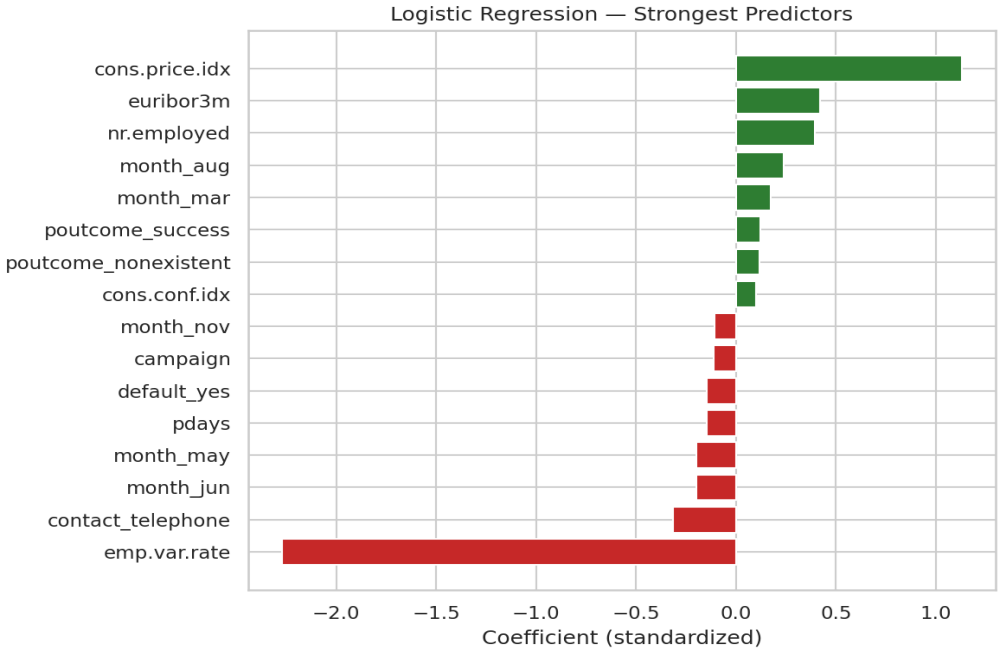


Figure 11. Strongest positive and negative predictors (standardized coefficients).

The model confirms the EDA: economic context and contact month dominate the strongest predictors, alongside previous-campaign success and contact channel. Model quality (accuracy ~83%, recall ~65%, ROC AUC ~0.80) is solid for a simple, fully interpretable baseline intended to rank and prioritize customers to call — not to serve as an automated final decision-maker.

8. Business Recommendations

- **Prioritize cellular contact** over telephone wherever a valid mobile number exists.
- **Re-engage previously successful customers first** in any new campaign — this segment converts at ~65%, by far the highest of any identified group.
- **Cap contact attempts at 2-3 per customer** per campaign; beyond that, returns diminish and may turn negative.
- **Sequence call lists by predicted conversion probability** from the logistic regression model, so agent time is spent on the customers most likely to convert.
- **Concentrate campaign volume in higher-converting months** (March, September, October, December) where scheduling allows, rather than defaulting to traditionally busy months like May.
- **Monitor macroeconomic indicators** (euribor rate, employment variation) as a leading signal for adjusting campaign scale and intensity.

9. Limitations

- The dataset covers one Portuguese bank's campaigns (2008-2010); findings may not generalize directly to other markets, products, or time periods.
- duration was deliberately excluded from the predictive model, so the model reflects what is knowable before a call takes place, not during one.
- Precision is modest (~0.36) due to class imbalance (~11% positive class); the model favors recall, which fits a "who should we call next" prioritization use case rather than a high-precision automated approval process.
- Logistic Regression assumes broadly linear/additive relationships; a more complex model could capture feature interactions, but was judged unnecessary added complexity for this project's business questions.

10. Conclusion

Campaign conversion is driven less by fixed customer demographics and more by factors within the bank's direct control: contact channel, timing, contact frequency, and how prior customer history is used. Combined with a simple, interpretable logistic regression model for ranking customers by likelihood to subscribe, the bank has a practical, low-complexity path to materially improving term-deposit campaign conversion without changing its core calling process.

Data source: UCI Machine Learning Repository — Bank Marketing dataset (https://archive.ics.uci.edu/dataset/222/bank+marketing). Original data collection and research by S. Moro, P. Cortez and P. Rita (2014). This report presents original analysis of that publicly available dataset for academic purposes.